Technical Brief: Classified Earthworks System

A Materials-Classification Architecture for Soil, Mulch, Biomass, Water, and Structured Land Deployment

Material-specific laboratory work, field trials, regulatory review, and qualified professional judgment remain necessary before any real-world release.

Author
Adam Ableman
Constitutional Computing
Contact: ableman.research@gmail.com

Abstract

This technical brief specifies a proposed classification and routing architecture for mixed soil, mineral, organic, water, and suspect material streams. It binds each output lot to source, sampling, testing, transformation, custody, intended use, and release standing so a material name cannot substitute for use-specific evidence. The system is a research and governance architecture, not a safety finding, engineering standard, environmental permit, agricultural recommendation, or authorization to deploy. Material-specific laboratory work, field trials, regulatory review, and qualified professional judgment remain necessary before any real-world release.

Status and Scope

This paper specifies a **proposed material-classification and routing architecture**. It is not an engineering standard, a contaminant-release protocol, an agricultural recommendation, a structural design, a waste-handling permit, or authorization to build or deploy any described process.

The architecture is intended to make material identity, source, sampling, testing, transformation, custody, intended use, and release standing inspectable. It does not establish that fired fines, pellets, biochar, soil blends, water works, structural earth, or remediation routes are safe or effective. Those claims require material-specific laboratory work, field trials, applicable regulatory review, and qualified environmental, agricultural, geotechnical, structural, hydrologic, fire, and occupational-safety judgment.

The governing claim ceiling is:

$$\mathrm{ReleaseStanding}(x, u) \preceq \mathrm{EvidenceStanding}(x, u).$$

No process, product name, or conceptual classification raises a material above the evidence held for its exact lot and intended use.

1. Executive Summary

The **Classified Earthworks System** is a proposed integrated processing architecture for transforming mixed earth, topsoil, subsoil, forestry residues, landscaping debris, construction spoil, and organic fines into tested, classified, purpose-built material streams.

Rather than treating "dirt," "mulch," "compost," "fill," and "waste" as separate categories, the system treats landscape material as a set of recoverable fractions. Each fraction is screened, washed, classified, tested, and routed according to its physical, chemical, biological, and structural suitability.

The system's core principle is:

Do not amend soil blindly. Classify each material fraction, then assign it to its proper future.

Clean mineral fines may be pelletized and fired into durable ceramic soil granules. Clean organic fines may be composted, pelletized, inoculated, or converted into biochar depending on the intended use. Coarse mineral fractions may become drainage media, terrace fill, structural fill, or aggregate. Coarse organic fractions may become mulch, erosion-control fiber, fungal substrate, or compost feedstock. Contaminated fractions are isolated for disposal, stabilization, or professional remediation.

The intended result is a governed research platform for evaluating whether classified material streams can support landscaping, forestry restoration, dryland remediation, construction-site recovery, terraced water management, garden soils, earth-building material preparation, and engineered root-zone design. Capability in any one use remains unestablished until the exact output and use profile are validated.

---

2. System Purpose

The Classified Earthworks System is designed to solve several overlapping problems:

1. **Loss of useful soil structure** Conventional screening, grading, excavation, and construction often destroy natural soil aggregation, leaving behind compacted, poorly draining, or biologically weak material.
2. **Misuse of fines** Clay, silt, compost fines, and organic dust contain mineral and nutrient value, but when left loose they can clog pore networks, reduce infiltration, create mud, or cause surface sealing.

3. **Unclassified contamination risk** Soil and fines from construction, urban, agricultural, roadside, or industrial sites may concentrate contaminants. Without classification and testing, potentially useful material may be wasted, or unsafe material may be reused improperly.
4. **Mismatch between material and deployment target** Garden soil, terrace fill, building material, mulch, forestry soil, drainage media, and water-harvesting basins require different particle distributions and performance profiles.
5. **Fragmented processing systems** Mulching, composting, soil washing, aggregate grading, biochar production, and earth-building material preparation are usually treated as separate processes. The Classified Earthworks System integrates them into one routing architecture.

---

3. Core Concept

The system receives mixed landscape or earthworks input and separates it into classified fractions:

- clean coarse mineral material,
- clean mineral fines,
- clean organic coarse material,
- clean organic fines,
- rocks and aggregate,
- wash water,
- contaminated or suspect fines,
- reusable amended soil blends.

Each fraction is then processed according to its intended deployment.

The initial architectural distinction is between **mineral fines** and **organic fines**. It is a routing hypothesis rather than an exhaustive material ontology; mixed, synthetic, treated, biological, and contaminated constituents may require additional classes.

3.1 Operative Primitives

The system becomes inspectable when its working nouns are explicit:

Primitive	Meaning
Feedstock lot	A bounded intake quantity with one retained identity, source history, custody record, and quarantine state.

Primitive	Meaning
Fraction	A physically separated portion of a lot. Separation creates a child identity; it does not erase parent history.
Sample	A custody-bound observation unit whose collection method and representativeness are recorded.
Test result	A result tied to a sample, method, laboratory or instrument, detection limits, date, and uncertainty.
Process profile	A declared transformation with admissible inputs, operating bounds, hazards, outputs, residues, and validation requirements.
Material class	A provisional descriptive class supported by current observations. It is not automatically a release class.
Intended-use profile	The exact agricultural, landscape, drainage, structural, thermal, disposal, or research use against which admissibility is evaluated.
Material passport	The governed record joining lot identity, ancestry, samples, tests, processes, custody, standing, restrictions, and revisions.
Release decision	A scoped decision that a particular output lot may enter a particular intended use under stated conditions.
Hold or refusal	A first-class outcome used when identity, evidence, authority, safety, or intended-use conditions are incomplete.

For a lot l, process p, and use u, release is not a property of the material name alone:

$$\begin{aligned}&\mathrm{Release}(l,p,u)\\&\Rightarrow I_l\\&\wedge S_{l,u}\\&\wedge T_{l,u}\\&\wedge A_{l,p,u}\\&\wedge C_l\\&\wedge V_{p,u}\end{aligned}\quad .$$

Here I_l is identity and ancestry continuity; $S_{l,u}$ is sampling adequacy for the intended use; $T_{l,u}$ is sufficient testing; $A_{l,p,u}$ is admissibility of the input-process-use combination; C_l is custody continuity; and $V_{p,u}$ is process and output validation. Failure or absence of a required conjunct yields hold or refusal, not optimistic reuse.

3.2 Safety and Standing Invariants

1. **Unknown is not clean.** An untested or inadequately identified lot remains suspect and quarantined.
2. **Use determines evidence.** Evidence adequate for one use does not automatically authorize another.
3. **Transformation is not erasure.** Washing, firing, composting, pyrolysis, pelletizing, dilution, or blending does not erase contaminants, uncertainty, or ancestry.
4. **Mixing cannot launder standing.** The output cannot receive a cleaner or safer standing merely because a suspect input was diluted into a larger lot.
5. **Fractions retain lineage.** Every child fraction remains traceable to parent lots, separation conditions, water streams, residues, and co-products.
6. **No release above evidence.** Product language and intended benefits remain hypotheses until the exact output and use are validated.
7. **Distinct consequence domains remain distinct.** Food production, child-contact landscapes, ecological restoration, stormwater treatment, load-bearing earth, buried water systems, and thermal processing require separate profiles and competent review.
8. **Residues count.** Process water, captured dust, ash, sediment, off-gas controls, rejected fractions, and contaminated equipment are part of the system boundary.
9. **Revision does not erase.** New tests may change standing while preserving the earlier record, its basis, and the reason for revision.
10. **No silent implementation authority.** A specification, recipe, model, or successful trial does not authorize deployment at another site or scale.

Mineral fines

Mineral fines include clay, silt, rock dust, weathered subsoil, bedrock grindings, and fine particles washed from topsoil or construction dirt. These fractions may carry useful mineral or water-retention properties, but their chemistry, contaminants, and behavior are site-specific. Loose fines may reduce pore continuity or contribute to sealing under some configurations; the direction and magnitude require measurement.

One research route proposes converting eligible mineral fines into **fired ceramic granules**. `Clean` here means admissible for the declared thermal process and intended use under a defined evidence profile; it is not a visual judgment. These

granules are designed to resemble irregular small-pebble particles: rough, porous, durable, and large enough to avoid behaving like dust.

Their role is to preserve the chemical value of fines while changing their geometry.

Organic fines

Organic fines include leaf dust, compost fines, ground twig fiber, bark fines, mulch slurry solids, and fine biomass from forestry or landscaping operations. These do not always need firing. In many uses, their value lies in decomposition, microbial feeding, fungal colonization, nutrient release, and humus formation.

Organic fines may become:

- compost pellets,
- mulch pellets,
- fungal nutrition clumps,
- seedbed amendments,
- erosion-control pellets,
- biochar feedstock,
- compost-biochar blends.

Their role is biological and nutritional rather than primarily structural.

---

4. System Principle

The governing principle is:

Keep the value of fines, but prevent loose fines from controlling the pore structure.

Fine particles are useful because they hold nutrients, water, and biological activity. They become problematic when they migrate, smear, seal, or clog. The Classified Earthworks System therefore returns fines to the landscape as **structured bodies**:

- mineral fines become ceramic crumbs,
- organic fines become nutrition pellets or compost aggregates,
- woody residues become mulch or fungal scaffolding,
- charred biomass becomes long-lived carbon storage and sorption media.

The design objective is to test whether drainage, aeration, water retention, fertility, and structure can be adjusted with more independence than in an unclassified blend. That separability is a hypothesis to be measured, not an established result.

---

5. Process Flow

The system can be organized into the following stages:

5.1 Intake

Input materials may include:

- construction dirt,
- screened topsoil,
- subsoil,
- excavation spoil,
- bedrock grindings,
- forestry slash,
- dropped limbs,
- twigs,
- leaves,
- mulch,
- compost,
- garden residues,
- landscape debris,
- clean demolition-adjacent soils, subject to testing.

Each load is logged by source, depth, location, date, visible condition, odor, known site history, and intended possible use.

5.2 Primary Screening

Material is mechanically separated into broad fractions:

- oversized rock,
- gravel,
- coarse mineral soil,
- sand-rich fraction,
- fine mineral fraction,
- coarse woody fraction,
- leafy/organic fraction,
- mixed fines.

5.3 Water Slurry Classification

Water is used to separate fines from coarser particles. This may include agitation, settling tanks, hydrocyclone separation, screens, or sedimentation basins.

Outputs include:

- washed coarse mineral fraction,
- suspended mineral fines,
- suspended organic fines,

- floating organics,
- sedimented fines,
- process water.

Water classification is valuable because it reveals the material that most strongly affects drainage, contaminant concentration, and soil behavior.

5.4 Testing Gate

Testing determines whether material may be reused, transformed, isolated, or rejected.

Test categories may include:

- particle-size distribution,
- organic matter content,
- pH,
- salinity/electrical conductivity,
- nutrient profile,
- cation exchange capacity,
- heavy metals,
- petroleum residues where relevant,
- pesticide residues where relevant,
- PFAS or industrial contaminants where relevant,
- pathogens for compost/manure-associated streams,
- shrink/swell behavior for structural soils,
- compaction performance for earthworks,
- infiltration/permeability for drainage media.

Testing is not an obstacle in the system. It is the routing mechanism.

5.5 Fraction Routing

After testing, fractions are routed into one of several material classes:

1. **Living soil stream** Clean topsoil, compost, leaf mold, charged biochar, soft organic pellets.
2. **Ceramic aggregate stream** Clean mineral fines shaped, dried, and fired into porous granules.
3. **Compost and nutrition stream** Clean organic fines composted, pelletized, inoculated, or blended.
4. **Biochar stream** Clean biomass pyrolyzed under low oxygen.
5. **Drainage and water-management stream** Stone, gravel, washed coarse mineral material, fired ceramic granules.
6. **Structural earth stream** Clean mineral soil blended for compaction, low organic content, and controlled clay/sand/gravel balance.

7. **Mulch and erosion-control stream** Coarse woody residues, bark, chips, fibers, fungal mats, surface-stabilizing material.

8. **Reject, regulated-management, or remediation stream** Contaminated fines, suspect sediments, treated wood residues, chemically unsuitable materials, high-salt fractions, unsafe ash, and any material lacking sufficient identity or evidence. Remediation is not authorized by classification; it requires its own competent authority and validation.

5.6 Governed Material State Machine

A material passport records the allowed state transitions:

```
INTAKE
  -> QUARANTINED
  -> IDENTIFIED
  -> SAMPLED
  -> CLASSIFIED
  -> TESTED_FOR(INTENDED_USE)
  -> ADMISSIBLE_FOR(PROCESS, INTENDED_USE)
  -> TRANSFORMED
  -> OUTPUT_SAMPLED
  -> OUTPUT_TESTED
  -> RELEASED_FOR(INTENDED_USE)
```

At every transition the system may instead emit `HOLD`, `REFUSE`, `REMEDIATE_UNDER_SEPARATE_AUTHORITY`, or `DISPOSE_UNDER_APPLICABLE_RULES`. A later intended use reopens admissibility; it does not inherit release merely because the same lot was accepted elsewhere.

The transition rule is:

$$\text{Admit}(s_i \to s_{i+1}) \iff \text{RequiredEvidence}(s_i, s_{i+1}) \subseteq \text{VerifiedEvidence}(l).$$

Each admitted transition produces a receipt binding the previous state, proposed transition, governing profile, evidence set, decision, actor or instrument, time, and resulting state. A receipt proves that a declared transition record exists; it does not by itself prove that sampling was representative, tests were correct, or the release was safe.

---

6. Material Products

6.1 Fired Mineral Granules

Clean mineral fines are shaped into irregular granules, dried, and fired to produce a durable porous ceramic aggregate.

Candidate design targets, not demonstrated properties:

- rough, irregular geometry,
- low dust,
- stable under wet/dry cycling,
- resistant to slaking,
- porous but not fragile,
- small-pebble particle range,
- compatible with soil blending,
- suitable for drainage and root-zone structure.

If validated, these granules would function as artificial soil aggregates. They are not fertilizer. Water, nutrient, pore-space, durability, leaching, dust, and biological effects require material- and use-specific testing.

Candidate uses requiring separate validation:

- garden soil structure,
- long-term landscape root zones,
- terraced berms,
- water-harvesting basins,
- tree pits,
- erosion-resistant soil blends,
- rain garden media,
- drainage-amended topsoil,
- restoration soils.

6.2 Organic Nutrition Pellets

Clean organic fines are compressed into small cylinders, BB-sized pellets, crumb pellets, or broken prills.

Target properties:

- biologically degradable,
- high surface area,
- moisture-absorbent,
- compatible with fungal colonization,
- easy to meter into soil blends,
- able to break apart gradually,
- optionally composted or inoculated before use.

Possible uses:

- forestry restoration,
- compost pile activation,
- garden fertility,
- fungal soil building,
- erosion-control berms,
- revegetation pockets,
- slow-release organic matter.

Fresh woody pellets should not be overused in annual crop root zones because high-carbon material can temporarily immobilize nitrogen. For gardens and farms, organic pellets should be composted, partially composted, nitrogen-balanced, or applied in layers rather than heavily mixed into the immediate root zone.

6.3 Biochar Pellets or Granules

Clean biomass may be pyrolyzed under low oxygen to produce biochar. Biochar may then be crushed, screened, pelletized, or blended with compost.

Target properties:

- stable carbon,
- porous structure,
- nutrient-holding potential,
- microbial habitat,
- low contaminant burden,
- appropriate pH and ash content,
- pre-charged before use where appropriate.

Possible uses:

- drought-resilient soils,
- nutrient retention,
- compost enhancement,
- long-term landscaping,
- degraded land restoration,
- desert berms,
- tree establishment,
- stormwater-adjacent soil blends.

Any conditioning or `charging` protocol, application rate, and productive-soil use must be defined and validated for the exact char, soil, crop or vegetation, exposure pathway, and intended outcome. The source architecture does not establish a generally safe application rule.

6.4 Coarse Mineral Drainage Fraction

Washed stone, gravel, coarse sand, and mineral aggregate can be used for:

- drainage layers,

- French drains,
- infiltration galleries,
- terrace-core drainage,
- cistern pre-filters,
- armored overflow paths,
- underdrains,
- roadbase-like support layers,
- erosion-resistant water pathways.

6.5 Structural Earth Fraction

Clean mineral soil may be blended for structural applications such as:

- compacted terrace fill,
- berm cores,
- rammed-earth candidate material,
- stabilized earth blocks,
- erosion-resistant landforms,
- non-food earthworks.

Structural earth should have low organic matter, predictable particle distribution, controlled moisture, and testing for compaction and strength. Organic pellets, compost, mulch, and high-biochar blends should not be used as load-bearing structural fill unless specifically engineered.

---

7. Deployment Profiles

The system does not produce one universal soil. It produces different blends for different targets.

7.1 Garden and Farming Profile

Purpose: fertility, safety, biological activity, root access, drainage, and water retention.

Likely components:

- clean screened topsoil,
- mature compost,
- charged biochar,
- moderate fired mineral granules,
- minimal fresh woody material,
- mineral amendments based on testing.

Design priorities:

- low contaminant burden,
- appropriate pH,

- balanced nutrients,
- stable aggregation,
- good infiltration,
- moderate water retention,
- minimal compaction.

7.2 Long-Term Landscaping Profile

Purpose: durable root-zone structure, reduced irrigation demand, stable drainage, and slow fertility.

Likely components:

- clean mineral soil,
- fired mineral granules,
- composted organic pellets,
- charged biochar,
- coarse woody surface mulch,
- optional mineral amendments.

Design priorities:

- long service life,
- resistance to settling,
- root-zone oxygen,
- moisture buffering,
- erosion resistance,
- compatibility with shrubs, trees, and perennial plantings.

7.3 Forestry and Revegetation Profile

Purpose: fungal succession, erosion resistance, seedling establishment, moisture retention, and long-term carbon building.

Likely components:

- clean mineral soil,
- woody mulch,
- composted fines,
- fungal inoculated pellets,
- biochar,
- coarse mineral drainage material as needed.

Design priorities:

- surface protection,
- fungal habitat,
- slow nutrient cycling,
- water absorption,
- low maintenance,

- compatibility with native vegetation.

7.4 Desert Hill and Terraced Berm Profile

Purpose: slow, spread, sink, store, and safely release water while establishing vegetation.

The profile should be layered rather than uniform.

Possible section:

- surface mulch or erosion blanket,
- root-zone blend with compost, biochar, and ceramic granules,
- water-storage layer with coarse mineral material and porous fired granules,
- compacted mineral terrace core,
- drainage layer,
- armored overflow path,
- cistern or infiltration-gallery connection where appropriate.

Design priorities:

- slope stability,
- controlled infiltration,
- erosion resistance,
- vegetation establishment,
- water harvesting,
- overflow safety,
- avoidance of buried decomposable structural voids.

7.5 Earth-Building and Rammed-Earth Profile

Purpose: strength, compaction, low shrinkage, low swelling, and durability.

Likely components:

- clean mineral soil,
- controlled clay fraction,
- sand,
- gravel,
- stabilizer where required,
- minimal organic material.

Design priorities:

- compressive strength,
- predictable shrinkage,
- low organic content,
- moisture control,
- particle-size optimization,
- structural testing,
- code and engineering review.

This profile is separate from garden soil. A good garden soil is often a poor structural earth material.

7.6 Drainage and Water-Management Profile

Purpose: move, store, filter, and release water.

Likely components:

- washed stone,
- gravel,
- coarse sand,
- fired ceramic granules,
- biochar where filtration or nutrient capture is desired,
- geotextile or separator layers where needed.

Design priorities:

- permeability,
- clogging resistance,
- clean inputs,
- stable grading,
- accessible maintenance,
- safe overflow routing.

---

8. Contamination and Remediation Logic

The system treats contamination testing as an opportunity rather than a nuisance.

Fine particles often carry a disproportionate share of contaminant risk because they have high surface area and can bind metals, hydrocarbons, and other pollutants. By separating fines from coarse fractions, the system can identify and isolate the riskiest material.

Possible outcomes:

1. **Clean fines** Routed to ceramic granules, soil blends, or structural mineral products.
2. **Nutrient-rich organic fines** Routed to composting, pelletizing, or biochar production.
3. **Contaminated fines** Removed, stabilized, encapsulated, or sent to professional remediation.
4. **Clean coarse mineral fraction** Reused as fill, drainage material, or aggregate after verification.
5. **Questionable or mixed material** Held in quarantine pending further testing.

The system should never casually fire, compost, pelletize, or distribute contaminated material. Thermal processing can concentrate, transform, or mobilize contaminants if not properly controlled.

---

9. Water Architecture

Water is both a processing tool and a deployment target.

During processing, water is used to slurry, separate, wash, and classify particles. In deployment, the resulting materials are used to control how water moves through landforms.

Potential water-management applications include:

- terraced berms,
- swales,
- rain gardens,
- infiltration basins,
- cistern-fed gravity irrigation,
- underdrains,
- qanat-inspired subsurface channels,
- spring interception systems,
- cave or bedrock-adjacent drainage works,
- erosion-resistant overflow systems.

The system is especially interesting where landform, hydrology, and soil restoration intersect: desert hills, degraded slopes, post-construction sites, wildfire-affected landscapes, forestry roads, and water-harvesting landscapes.

Subsurface water systems, tunnels, cisterns, cave-adjacent works, and qanat-like designs require professional geotechnical, hydrologic, structural, and safety review.

---

10. Design Philosophy

The Classified Earthworks System is not merely a composting system or soil amendment process. It is a material-intelligence system for landscapes.

Its philosophy can be summarized as follows:

- Soil is not one material.
- Fines are not waste.
- Contamination is not guessed; it is tested.
- Clean fractions are routed by use.
- Dirty fractions are isolated.
- Structure and fertility are designed separately.

- Water movement is designed deliberately.
- Organic material feeds life but should not carry structural loads.
- Mineral material can carry structure but may need biological enrichment.
- Different land uses require different particle bodies.

The system does not ask, "How do we make this dirt good?"

It asks:

What is each fraction fit to become?

---

11. Corpus Integration and the Governed Meaning Base

The author-confirmed terminology lineage is `CCF -> CMB -> GMB`. **Governed Meaning Base (GMB)** is the current name. CCF and CMB remain historical names in exact titles, quotations, source descriptions, and lineage explanations; they are not separate current architectures.

For Classified Earthworks, GMB is the governed semantic substrate for material passports. It can retain lots, fractions, ancestry, samples, methods, tests, process profiles, intended uses, restrictions, decisions, receipts, disputes, and revisions. Recording a claim in GMB does not make the claim true and does not authorize physical release.

The direct corpus relations are:

Corpus work or structure	Relation to Classified Earthworks
Semantic Legitimacy Before Context	A material description becomes actionable context only after identity, admissibility, standing, and intended use are explicit.
Reasoning by Way of Receipts	Each sampling, classification, process, test, hold, refusal, release, and revision can be reconstructed as a bounded transition rather than an unexplained conclusion.
The Three Hash Rule	Separates the exact lot or output state, the admissibility and process configuration, and the meaning or release commitment so one cannot silently substitute for another.
A Formal Theory of Work Custody	Tracks who bears the live duty for sampling, quarantine, transformation, release review, and exception resolution without treating custody as proof of correctness.

Corpus work or structure	Relation to Classified Earthworks
Admissibility Engineering	Converts the contamination gate and use-specific release rule into schemas, mediators, refusal paths, and bypass tests.
GMB	Preserves the meaning and evidence fabric of the material passport while keeping author claims, AI synthesis, laboratory evidence, and release authority distinct.

These are typed interfaces, not automatic work or system identity mergers. The material architecture is a domain realization candidate of the wider governance grammar, not evidence that it has already been implemented or validated.

12. Realization Funnel

Stage	Classified Earthworks content
Possibility	Mixed earth and biomass may be treated as recoverable, evidence-bearing material fractions rather than undifferentiated waste.
Specification	Define lot identity, fractions, sampling, tests, process profiles, intended uses, material passports, release standing, holds, refusals, residues, and revision.
CM1	Enumerate admissible classification, processing, reuse, remediation, disposal, and non-action configurations, including brief, harmful, or undesirable possibilities that must remain visible for analysis.
CM2	Prefer configurations that protect people and ecosystems, minimize virgin inputs and waste only where demonstrated, preserve custody, expose uncertainty, prevent contamination laundering, and remain inspectable and reversible.

Stage	Classified Earthworks content
Instance Configuration	Select site, feedstock classes, jurisdiction, intended uses, sampling plans, laboratories, thresholds, process bounds, equipment, qualified reviewers, emergency rules, and maintenance responsibilities.
Implementation Realization	Build a bounded pilot yard, passport store, quarantine layout, sampling workflow, process cells, residue controls, release gate, and monitored demonstration sites.
Evidence	Bind exact lots, samples, methods, results, process telemetry, output tests, field observations, adverse events, comparisons, limitations, and independent review.
Revision	Change profiles or standing in response to new evidence while preserving prior states, reasons, affected lots, and downstream recall obligations.

The present paper primarily contributes Specification and CM1. Its safety preferences and routing philosophy contribute CM2 candidates. The implementation phases below are research configurations, not permission to construct, process, release, or distribute materials.

13. Implementation Research Phases

Research Phase 1: Pilot Classification Yard

Establish a small processing yard with:

- intake logging,
- screening,
- water classification,
- drying beds,
- test sampling,
- clean/suspect/contaminated storage zones,
- basic soil blending,
- composting area,
- organic pellet trial,
- mineral granule trial.

Primary objective: prove classification, routing, and safe reuse logic.

Research Phase 2: Material Product Development

Develop repeatable material products:

- fired mineral granule,
- compost pellet,
- mulch pellet,
- biochar pellet,
- garden soil blend,
- landscape soil blend,
- terraced berm blend,
- drainage blend,
- structural earth candidate blend.

Primary objective: produce tested recipes with measurable performance.

Research Phase 3: Deployment Trials

Install controlled demonstration sites:

- raised garden bed,
- tree planting pit,
- rain garden,
- terraced berm,
- slope restoration area,
- forestry revegetation strip,
- compacted structural fill test pad,
- drainage/infiltration test cell.

Primary objective: compare infiltration, plant growth, erosion resistance, settlement, and maintenance performance.

Research Phase 4: Integrated Earthworks Platform

Scale to mobile or semi-mobile operations for:

- construction sites,
- municipal green-waste yards,
- forestry operations,
- land restoration projects,
- desert water-harvesting landscapes,
- regenerative developments,
- earth-building material preparation.

Primary objective: convert earth-moving waste streams into classified deployment materials.

Reader-Facing Realization Brief

Across the completed P1 corpus, this Technical Brief is the **reader-facing realization brief for the broader Classified Earthworks research architecture**. It should present the system's integrated process, material routes, site profiles, risks, and validation program without becoming a second foundational identity or implying that a proposed route is safe, permitted, or proven.

- **Adjacent work:** *Classified Earthworks*
  Supplies: The parent classification, routing, custody, and material-state architecture.
  Brief boundary: This brief compresses the parent for technical orientation and staged investigation.
- **Adjacent work:** *Technical Brief: Hydroengineering and Geologic Intelligence Modules*
  Supplies: Site-water and geologic-intelligence specialization.
  Brief boundary: Those modules remain profiles under the parent, not independent authorization for infiltration or excavation.
- **Adjacent work:** *Semantic Legitimacy Before Context*
  Supplies: Identity, admissibility, standing, and intended-use checks before a fact becomes actionable context.
  Brief boundary: The material passport must not turn a measurement into an unrestricted release claim.
- **Adjacent work:** *A Formal Theory of Work Custody*
  Supplies: Live responsibility for lots, samples, holds, transformations, reviews, placement, and recall.
  Brief boundary: Custody continuity supports actionability but does not establish material suitability.
- **Adjacent work:** *Reasoning by Way of Receipts*
  Supplies: Reconstructable classification, process, release, refusal, and revision transitions.
  Brief boundary: A receipt proves only its bounded record and participating evidence.

The brief should organize every proposed material route through a route card:

$$\begin{aligned} R = (&input, identity, hazards, tests, process, output, \\ &intended_use, conditions, refusals, residues, authority,. \\ &evidence, monitoring, recall) \end{aligned}$$

No route moves directly from a material name to a product claim. Each route has a use-specific decision and may end in `UNKNOWN`, `QUARANTINE`, `MORE_EVIDENCE`, `PILOT_ONLY`, `CONDITIONED_USE`, `REFUSE`, or

ELIGIBLE_FOR_SEPARATE_RELEASE_REVIEW. Reuse, diversion, carbon, water, ecological, and cost benefits remain independent claim families.

The realization funnel in the brief should retain the now-author-confirmed terminology: IC may mean Instance Constitution or Instance Configuration, and realization-stage IR may mean Instance Realization or Implementation Realization. These interchangeable expansions do not authorize construction. The physical pilot remains a candidate IR only after site, authority, professional review, safety, environmental, and evidence obligations are bound in a specific IC.

The universal GSM can type
PROFILES_CLASSIFIED_EARTHWORKS, ROUTES_MATERIAL_FRACTION,
PROPOSES_USE_SPECIFIC_PROFILE, HOLDS_LOT_IN_QUARANTINE, GENERATES_RESIDUE_STREAM,
REQUIRES_SEPARATE_RELEASE_REVIEW, MONITORS_PLACED_MATERIAL, TRIGGERS_RECALL,
FAILS_SYSTEM_BENEFIT_CLAIM,
and
REMAINS_DISTINCT_FROM_PARENT_WORK. These relations keep the accessible brief connected to the deeper research without duplicating it.

This integration establishes four editorial commitments: designate the document as the reader-facing realization brief; adopt the route-card schema and explicit route outcomes; keep every physical-use and benefit claim use-specific and separately evidenced; and retain the parent paper, hydro/geologic module brief, and this brief as distinct but linked publication roles.

Physical Route Evidence and Cross-Domain Nonrelation

The fixed-seed comparison with Warrant by Structure, Context-Compiled Consciousness Mode, Digital Gold as a Frozen Compiler, and Dynatology tests whether the brief keeps physical evidence distinct from architectural analogy:

$$\begin{aligned} R^* = (&material_identity, site, route, hazards, tests,\\ &authority, process, placement, monitoring, outcome,.\\ &warrant, recall, revision) \end{aligned}$$

Structural warrant must be use- and site-specific. Laboratory classification, process receipts, professional review, placement records, and monitoring support different claims. No document structure or software-style mediation substitutes for contamination, geotechnical, hydrologic, environmental, occupational, or longitudinal field evidence.

Context-compiled reflective modes and monetary automata have no material evidentiary relation to soil, biomass, water, or land deployment. Shared compiler or realization language is insufficient. These nonrelations are recorded so the universal GSM does not transform metaphor into system identity or evidence transfer.

Dynatology can describe possible, admissible, preferred, selected, realized, observed, viable, and revised route states only through a domain-specific mapping that preserves physical constraints and observations. That grammar cannot establish which material route is safe or lawful.

The universal GSM can type
`WARRANTS_PHYSICAL_ROUTE_CLAIM`,
`REQUIRES_SITE_SPECIFIC_EVIDENCE`, `MONITORS_PLACEMENT_OUTCOME`,
`RECALLS_MATERIAL_ROUTE`,
`MAPS_ROUTE_TO_DYNATOLOGICAL_SYSTEM`, `REFUSES_CROSS_DOMAIN_EVIDENCE_TRANSFER`,
and
`REVISES_AFTER_FIELD_RESULT`.

This randomized review establishes four editorial commitments: adopt the expanded physical route object; require site-specific warrant; record two cross-domain nonrelations; and use Dynatology only through a preservation-explicit domain mapping.

Material Passport, Release, and Recall Profile

Three downstream uses establish this brief as the reader-facing system projection of the deeper *Classified Earthworks* model, a component of the hydroengineering/geologic profile, and a domain case for Warrant by Structure. The paper should therefore center an exact material-lot passport and governed release case rather than imply that a general constitutional vocabulary validates earthwork practice.

$$\begin{aligned} M^* = (&lot, source, chain_of_custody, sampling, methods,\\ &observations, hazards, classifications, uncertainty,\\ &transformations, destination, release_criteria,\\ &authority, witnesses, monitoring, recall, expiry,\\ &remainder) \end{aligned} \quad .$$

Source class, measured composition, contamination status, treatment state, intended use, and release decision are distinct. A sample result has spatial, temporal, method, detection-limit, and chain-of-custody bounds. Classification cannot erase unknown contaminants, heterogeneous material, hydrologic interaction, biological change, or future exposure. A model recommendation is not a release authority, and a passport is not proof that the represented lot still matches the sampled lot.

Every transformation must preserve predecessor identity, mass or volume reconciliation where feasible, introduced materials, process conditions, changed hazards, rejected fractions, and downstream monitoring obligations. Recall and remediation need destination traceability and responsible custody. Domain experts and applicable law remain outside the authority of this candidate brief.

The universal GSM can type
`IDENTIFIES_MATERIAL_LOT`, `SAMPLES_UNDER_METHOD`,
`CLASSIFIES_WITH_UNCERTAINTY`, `TRANSFORMS_MATERIAL_LOT`, `AUTHORIZES_RELEASE_SEPARATELY`,
`MONITORS_DEPLOYED_MATERIAL`, `RECALLS_BY_PASSPORT`, and
`RETAINS_PHYSICAL_REMAINDER`.

This reverse-dependency review establishes four editorial commitments: classify the brief as the reader-facing system projection; adopt the material passport and release case; bind sampling, uncertainty, transformation, destination, monitoring, and recall; and keep technical recommendation, legal authority, and physical safety evidence separate.

14. Key Risks

The major risks are:

- reuse of contaminated material,
- inadequate testing,
- poor separation between clean and suspect streams,
- firing contaminated fines,
- using organic-rich material in structural fills,
- excessive fresh woody material in crop root zones,
- clogging caused by unstable pellets or dust,
- slope failure from over-infiltration,
- uncontrolled runoff,
- poor compaction design,
- salinity buildup,
- pH imbalance,
- water-rights or permitting issues,
- unsafe excavation or subsurface construction.

These risks are manageable only if the system is treated as a technical platform, not as a casual landscaping gadget.

---

15. Evidence and Validation Program

The system should be evaluated as linked claims rather than as one undifferentiated success claim.

Claim family	Minimum evidence direction
Classification	Repeatable separation, mass balance, particle distributions, fraction identity, water and residue accounting, and uncertainty.

Claim family	Minimum evidence direction
Contamination control	Use-specific analyte plan, qualified sampling, detection limits, laboratory custody, blank/duplicate controls where applicable, and disposition of every residue stream.
Fired mineral granules	Feedstock chemistry, firing profile, emissions and residues, strength, slaking, abrasion, dust, leaching, wet/dry and freeze/thaw behavior, and use-specific field performance.
Organic pellets or compost products	Feedstock exclusions, process conditions, maturity or stability criteria, pathogens where relevant, salinity, nutrient behavior, decomposition, oxygen effects, and field trials.
Biochar products	Feedstock identity, pyrolysis conditions, contaminants, pH, ash, stability, dust and fire hazards, application profile, and field response.
Structural earth	Geotechnical classification, moisture-density relation, strength, shrink/swell, settlement, drainage, durability, and qualified design review.
Hydrologic works	Site water balance, infiltration and clogging behavior, slope stability, overflow and failure routing, downstream effects, maintenance, and applicable rights or permits.
Agricultural or ecological use	Soil and crop or vegetation response, contaminant exposure pathways, nutrient effects, salinity, persistence, ecological side effects, and comparison against a declared baseline.
System-level benefit	Complete energy, water, labor, emissions, waste, virgin-material, cost, and risk accounting rather than benefits inferred from diversion alone.

Negative results, holds, unusable fractions, process failures, off-specification outputs, and adverse effects are first-class evidence. A pilot succeeds epistemically when it makes its limits and refusal conditions clearer, even when a proposed product route fails.

16. Research and Development Priorities

Important R&D questions include:

1. What firing temperatures create durable but still porous mineral granules from local fines?
2. Which granule sizes best improve drainage without reducing water retention?
3. How do ceramic granules perform after repeated wet/dry and freeze/thaw cycles?
4. How much mineral fines value is preserved after firing?
5. What organic pellet formulations provide nutrition without creating anaerobic pockets?
6. When should organic fines become compost versus pellets versus biochar?
7. How should biochar be charged before deployment?
8. What testing protocol is sufficient for different reuse classes?
9. How should contaminated fines be isolated and handled?
10. What soil recipes work best for gardens, terraces, forestry, rain gardens, and structural fill?
11. How do terraced berm blends perform under intense rainfall?
12. Can the system reduce import of virgin aggregate, compost, and engineered soil media?
13. Can the system safely support water-harvesting landscapes in dry regions?
14. Can classified mineral streams support rammed-earth or stabilized-earth construction?

A Material Passport Is Not a Safety Certificate

Public orientation

The Classified Earthworks System proposes a governed way to identify, sample, transform, route, monitor, and recall soil, biomass, water, aggregate, and other heterogeneous material lots. Its strongest contribution is a material-passport and release architecture. It is not a construction plan, agricultural prescription, remediation method, waste permit, product certification, or finding that any material is safe.

Adversarial tests

1. **Sampling-substitution attack.** A small or convenient sample is treated as the represented lot despite spatial, temporal, depth, moisture, and handling heterogeneity.
2. **Unknown-contaminant attack.** Passing a selected panel is interpreted as clean even though untested chemicals, biological agents, debris, fibers, treatment residues, or transformation products may remain.

3. **Transformation-identity attack.** Washing, heating, composting, pelletizing, blending, or storage changes hazards and performance while the predecessor classification silently persists.
4. **Destination-mismatch attack.** A result valid for one use, exposure route, jurisdiction, site, climate, or population is reused for another.
5. **Mass-balance attack.** Rejected fines, wash water, emissions, leachate, dust, oversize material, and losses disappear from the account while useful outputs receive full custody.
6. **Passport-certificate attack.** A complete data record is mistaken for professional approval, regulatory compliance, engineering fitness, or release authority.
7. **Recall-gap attack.** Material is split, mixed, sold, deployed, or transported beyond the traceability needed for recall, containment, or remediation.
8. **Community-exposure attack.** Technical classification proceeds without affected-community, Indigenous-data, labor, land-use, ecological, or downstream-water obligations.

Claim and evidence ceiling

The paper specifies a materials-classification and routing proposal with custody, sampling, transformation, release, monitoring, and recall requirements. It does not establish that any named input, process, output, product line, facility, or deployment is safe, lawful, effective, economical, sustainable, or ready to build.

Every lot and use requires declared sampling bounds, verified methods, chain of custody, mass and residual accounting, site-specific exposure and performance evidence, qualified domain judgment, and applicable approvals. No testing, handling, construction, deployment, or external research was authorized or performed in this pass.

Reader-facing consequence

The public value is a way to keep physical material claims attached to the exact lot, tests, transformations, destination, authority, monitoring, and recall path - while refusing to turn data organization into permission to use the material.

17. Publication Standing and Evidence Limits

17.1 Current Standing

This candidate is a conceptual system architecture and research program. The present review found no exact experimental dataset, laboratory report, engineering analysis, field-trial package, regulatory opinion, or independently reproduced implementation evidence attached to this paper. Product properties, environmental benefits, safe reuse routes, and system-level outcomes therefore remain hypotheses or design targets.

The paper may be publication-ready as a clearly labeled proposal if its claim ceiling remains explicit. It is not yet ready to function as a technical standard, validated product specification, remediation protocol, structural design, or operational guide. No external research was performed during this closed-corpus pass; environmental, agricultural, engineering, safety, standards, and related-work claims require later primary-source verification.

PDF rendering remains deferred until textual and semantic saturation. The authorized baseline is ordinary Pandoc mode only:

```
pandoc INPUT.md -o OUTPUT.pdf
```

17.3 Corpus Sources

- Adam Ableman, *Classified Earthworks System.* Exact author source SHA-256 `539C18D05AB84BBE216681D010E189E6495EE4AB1BE9FF541033F5D05FDF8A55`.
- Adam Ableman, *Reasoning by Way of Receipts.* Exact reviewed source SHA-256 `8D884B96597EA6303A324173A7CE2D8C735A952E91FBE4C9EB6509FA6D6B195F`.
- Adam Ableman, *A Formal Theory of Work Custody.* Exact reviewed source SHA-256 `49C38F60B95E107F8B3D864ACAFB4C49067A146C807E5248307DE7B009AD4696`.
- Adam Ableman, *Admissibility Engineering.* Exact reviewed source SHA-256 `FA80D5895A569839958193735B754BE62EE20E1B399A06CB3BA959614DEBF7A3`.
- Adam Ableman, *Semantic Legitimacy Before Context.* Exact reviewed source SHA-256 `1E9058F212FCB00E42E35E927DF2E7F830BC9C9499AAFF469FE65FF8AE137B68`.
- Adam Ableman, *The Three Hash Rule.* Zenodo, 2026. DOI: `10.5281/zenodo.20788617`; exact reviewed source SHA-256 `03F60317CC9AFB36DDBCF9BF1C6D185611666CC4D9AB290BD4BCC17DF4579B7C`.
- Governed Meaning Base terminology ruling: `CCF -> CMB -> GMB`, with GMB current. This ruling establishes terminology lineage only.

The cross-work relations, formalization, realization-funnel placement, evidence program, and publication standing in this candidate are AI-prepared editorial synthesis explicitly distinguished from the original author material. They grant no publication, canonical, engineering, regulatory, implementation, or deployment standing.

18. Conclusion

The Classified Earthworks System proposes a shift from bulk earthmoving to intelligent material routing.

It takes mixed earth and biomass and separates them into useful fractions. It proposes routes by which eligible mineral fines might become ceramic granules, eligible organic fines might become tested biological products, identified woody

residues might become mulch or process feedstock, eligible biomass might become characterized biochar, coarse mineral fractions might enter validated drainage or structural profiles, and suspect fractions might remain quarantined or enter separately authorized management.

The system is especially promising for disturbed sites where soil has already been broken: construction spoil, compacted subsoil, degraded slopes, forestry residues, desert hills, post-fire landscapes, urban fill, and restoration earthworks.

Its highest-value application is not simply making compost or topsoil. Its highest-value application is building landscapes that know how to hold water, breathe, grow roots, resist erosion, and route each material according to its actual properties.

In this framework, a hill of dirt is not waste.

It is an unclassified library of particles.

The work is to read it.

Corpus-Wide Reference Positioning

This paper argues that this technical brief specifies a proposed classification and routing architecture for mixed soil, mineral, organic, water, and suspect material streams. Its nearest external comparison fields are earth, cps. The sources in [E1], [E2], [E3], [E4], [E5], [E6], [E7] are cited as predecessor literature, adjacent standards, or counterpressure - not as evidence that they caused the Ableman framework or that they validate its novel claims. The comparison isolates where established work already supplies provenance, formal assurance, institutional constraint, empirical method, or engineering practice, and where this paper adds a distinct claim about admissibility, authority, semantic custody, or governed realization.

Within the Ableman corpus, the highest-priority current cross-links are [A1], [A2], [A3], [A4], [A5], [A6], [A7]. These are links among separately custodied works, not silent identity mergers. Their role is to expose mutual dependencies, terminology collisions, and places where a reader should traverse the corpus before treating this paper as self-sufficient.

Reference-pass counterevidence and limit search

The pass applied the following pressure tests: Material classes are sampling- and method-dependent; heterogeneous sites, contaminant transport, and regulatory thresholds can defeat a neat classification-to-action mapping. Runtime monitors and fallback controllers only protect modeled safety envelopes and can fail through sensor error, timing assumptions, unsafe recovery, or omitted hazards. These limits are not generic disclaimers: they identify the assumptions that a later empirical, formal, or implementation review must try to break.

Reference-pass influence and novelty boundary

No newly added external source is asserted to be a documented authorial influence unless the predecessor source already said so. `Predecessor` means earlier relevant work;

`adjacent` means a useful comparison; `standard` means an interoperability or assurance baseline; and `counterevidence` marks work that narrows or pressures a claim. Novelty therefore remains a reviewable, claim-level question rather than an inference from a low similarity score or an absent citation.

References

External scholarship, standards, and prior art

[**E1**] U.S. Army Corps of Engineers. *Sediment Investigations*, Engineer Manual EM 1110-2-4000. 14 February 2025. publications.usace.army.mil source **Relationship:** Primary engineering guidance for material characterization and classification; included to ground classification claims in field and laboratory practice.

[**E2**] U.S. Environmental Protection Agency. *Guidance for Cleaning Up Groundwater, Soil and Air at Hazardous Waste Cleanup Facilities.* epa.gov source **Relationship:** Primary remediation guidance emphasizing media transfer, exposure, and site-specific cleanup decisions.

[**E3**] Edward R. Griffor et al. *Framework for Cyber-Physical Systems: Volume 1, Overview.* NIST SP 1500-201, 2017. DOI: 10.6028/NIST.SP.1500-201. https://doi.org/10.6028/NIST.SP.1500-201 **Relationship:** Adjacent cyber-physical systems framework for trustworthiness, timing, data, and lifecycle concerns.

[**E4**] W3C. *PROV-O: The PROV Ontology.* W3C Recommendation, 30 April 2013. https://www.w3.org/TR/prov-o/ **Relationship:** Adjacent provenance standard; it models entities, activities, and agents, but does not by itself establish the paper's stronger authority or standing claims.

[**E5**] J. Tanner Slagel et al. "A Formal Verification Framework for Runtime Assurance." *NASA Formal Methods*, 2024. NASA NTRS 20230017350. https://ntrs.nasa.gov/citations/20230017350 **Relationship:** Adjacent runtime-assurance formalization, useful for comparing failover safety with constitutional admission and effect recognition.

[**E6**] Nancy G. Leveson and John P. Thomas. *STPA Handbook.* MIT, 2018. https://psas.scripts.mit.edu/home/books-and-handbooks/ **Relationship:** Adjacent systems-theoretic hazard analysis method; used to test treatment of unsafe control and system-level constraints.

[**E7**] Robert N. M. Watson, Simon W. Moore, Peter Sewell, and Peter G. Neumann. *An Introduction to CHERI.* University of Cambridge Technical

Report UCAM-CL-TR-941, 2019. DOI: 10.48456/tr-941. cl.cam.ac.uk source **Relationship:** Adjacent capability architecture demonstrating unforgeable hardware authority and compartmentalization.

Ableman corpus

[**A1**] Adam Ableman. *Classified Earthworks.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-8D3F34F396D7760E`;
source SHA-256
`9011C1800E2294A0F0E72BFBA871B071C8C4590C2A7E83FEBFF2BAE5D29E533A`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A2**] Adam Ableman. *Technical Brief: Hydroengineering and Geologic Intelligence Modules.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-066CA7D56F2416A8`;
source SHA-256
`396731CC5A67026E88FD6028CD6A553066F7F7C352FA8072CC44389EBB93E94A`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A3**] Adam Ableman. *Water: MCIR as the Engineering of Governed Change.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-91A910961BA1B5AF`;
source SHA-256
`1C0B814A76E0724FAD8ECC7A71DBBFC55B3C13E8ED03ED6F399496A70C2587CD`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A4**] Adam Ableman. *Constitutional Robotics.* Ableman Corpus current edition; work identity `PAPER.PUB.18750683`;
source SHA-256
`D10DF9B6B7F742D9EEABF01BC5DFC3755BB4EBF548E8CB5F215BB9D94678F514`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A5**] Adam Ableman. *A Constitutional Syscall Architecture for Cyber-Physical Control Systems.* Ableman Corpus current edition; work identity
`AMCS-LOCAL-WORK-F46D86D479F89BAC`;
source SHA-256
`5E696B753A10BD3C3D36B675A84B95CCF2234F024A6B277BF121E76BFCBCBD18`.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A6**] Adam Ableman. *Effect Sovereignty.* Ableman Corpus current edition; work identity `PAPER.PUB.18737323`;
source SHA-256

1DB8A34946A8705F6F4BA4FC3B12B251FC7C83DBDF217F9DDFF55529C8612EE7.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

[**A7**] Adam Ableman. *Constitutionally Governed Infrastructure.* Ableman Corpus current edition; work identity
AMCS-LOCAL-WORK-EF5747107D56F516;
source SHA-256
9C708BA381108EEA252404DF33F7B3B1648EC9CF5F2ACF00BAB88C6A15540926.
Corpus relationship: priority cross-link selected by corpus profile and similarity review; no identity merger or publication effect.

Editorial source notes retained from predecessor

The entries below are corpus-internal comparison and dependency sources. Their inclusion does not establish external validation of this paper's empirical, safety, novelty, or performance claims.

Adam Ableman *The Three Hash Rule.* Zenodo, 2026. doi:10.5281/zenodo.20788617

Adam Ableman *Constitutional Computation.* Zenodo, 2026. doi:10.5281/zenodo.20624488

Adam Ableman *Constraint-Bearing Composability.* Zenodo, 2026. doi:10.5281/zenodo.21632061

Adam Ableman *Fractal Governance Architecture.* Zenodo, 2026. doi:10.5281/zenodo.20754189

Adam Ableman *MCIR: A Canonical Intermediate Representation for Governed Transformation.* Zenodo, 2026. doi:10.5281/zenodo.21095303

Adam Ableman *Semantic Coordinate Identity Tokenization (SCIT).* Zenodo, 2026. doi:10.5281/zenodo.20532070

Adam Ableman *Semantic Memory Hierarchies.* Zenodo, 2026. doi:10.5281/zenodo.20793071

Corpus source and companion formats: https://github.com/AblemanMosaic/ableman-corpus
License: CC-BY-ND-4.0

DOI: 10.5281/zenodo.23847486